

Multiplication with a Reasonableness Check



Grade 5 • Mathematics
NC.5.NBT.5

Use multi-digit multiplication accurately.



What will a strong explanation show?

1. A model or equation that fits the quantities.
2. Accurate calculation with appropriate units.
3. A check that makes the answer reasonable.



Watch the reasoning

The community center packs 107 cards in each of 22 boxes. Find the total. Show partial products and explain why they can be added.

$107 \times 20 + 107 \times 2 = 2140 + 214 = 2354$. The parts account for every box or card exactly once.



Build a second model together

The science club packs 148 cards in each of 12 boxes. Find the total. Show partial products and explain why they can be added.

Discuss: What should the first step find?



Your turn • 1

The science club packs 15 cards in each of 12 boxes. Find the total. Show partial products and explain why they can be added.

Represent → solve → explain → check



Your turn • 2

The science club packs 63 cards in each of 29 boxes. Find the total. Show partial products and explain why they can be added.

Represent → solve → explain → check



Your turn • 3

The music club packs 363 cards in each of 26 boxes. Find the total. Show partial products and explain why they can be added.

Represent → solve → explain → check



Your turn • 4

The music club packs 269 cards in each of 15 boxes. Find the total. Show partial products and explain why they can be added.

Represent → solve → explain → check



Exit question • Explain independently

The library team packs 329 cards in each of 19 boxes. Find the total. Show partial products and explain why they can be added.

Include a check. What would you teach someone else?



Teacher review and next steps

1. 180

Exit: 6251

2. 1827

Use the worked PDF key and slide notes to discuss methods. Revisit the earliest unclear step before adding more practice.

3. 9438

4. 4035

